Cloud-based ambulance tracking and movement facilitation through traffic light control

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***Abstract*— The proliferation of roads and the high number of cars present serious problems for transportation networks today. Ambulance services are among the numerous services that are most affected by traffic congestion. This could put the health of the person inside the ambulance at risk. This problem can be resolved by using the Control System for Smart Traffic Signals with LTE, GPS, and RFID for Ambulance, which links the traffic light and the ambulance. RFID (radio frequency identification) technology is used to implement intelligent traffic signal control. The main concept of the suggested system is that if a signal light stops an ambulance, RFID diffused at the traffic would follow it and transmit the data to the cloud. The light changes back to green once the ambulance passes by, and the signaling pattern resumes as it did before. This system will find the ambulance and control the traffic lights if it is completely automated. All of these events occur without the need for human intervention. This procedure continues until the ambulance arrives at the hospital. It is a time-saving project that may be used to control traffic lights in an emergency.**

Keywords*— *RFID, cloud, ambulance, traffic lights.

I. Introduction

Due to the city's quick expansion, the size of the city is growing every day, and its population is soaring [1]. One of the most important concerns in cities around the world is traffic congestion [2]. The most important goal is to be at the hospital on time and avoid traffic jams. Diverse technologies, like as image processing, are used in various techniques for monitoring and managing traffic [3], [4]. magnetic detectors [5], infrared detectors [6], global positioning system (GPS) [7], and Radio Frequency Identification (RFID) [8], [9]. The Global Positioning System (GPS) is made up of more than 20 satellites that allow for precise position triangulation. [10]. GPS technology is mostly used in automobiles to make route planning and vehicle location estimates easier. Direct line-of-sight is the most important criterion for accurate GPS triangulation. All GPS signals can be effectively blocked by contemporary design, such as densely populated cities and underground tunnels, implying that accurate position triangulation is impossible. Multi-path effects may be common in crowded urban environments because of the high-rise structures there [11]. In addition to the limitations described above, precise vehicle location requires a current map database [11]. To get beyond the restrictions of a solo GPS system, many technologies can be integrated with GPS technology, such as GSM and RFID.

RFID is a low-cost and effective method of tracking stocks and other moving objects [2]. As a result, RFID technology should be taken into account while designing intelligent traffic monitoring systems. RFID technology's advantages can be observed in present applications. The benefit of an RFID transponder is that it doesn't need line-of-sight and, more importantly, doesn't require human interaction [12]. This enables for improved time management and usage of human resources.

In recent years, RFID technology has advanced from obscurity to widespread application, enabling quicker processing of manufactured items and materials. Unlike earlier bar-code technology, which required a line of sight, RFID enables identification at a distance. In addition to storing additional data like the product's maker, category, and even information that is related to environment like temperature, RFID tags have the potential to record a higher number of individual IDs than bar codes [12].

Embedded technology is used in the prior existing system proposed in [13] for Intelligent Traffic Control using GSM and GPS. While the ambulance is stuck in traffic, the fundamental goal of this article is to turn the opposing traffic signal board to red and reset the timer, which is the traffic signal board, to green.

The researcher in [14] devised and implemented a GPS-GPRS-based vehicle tracking system, while the research [15] recommends a system depending on a wireless sensor network as well as a warning system based on the line length line of parked cars next to traffic posts for red light crossing scenarios.

In this study, the basic idea behind the proposed system is that RFIDs dispersed throughout the traffic would track an ambulance in the event that a signal light stopped it and send the information to the cloud. After the ambulance passes by, the light turns green again, and the signaling pattern returns to its previous state. If this system is fully automated, it will locate the ambulance and manage the traffic signals without human interaction.

The remainder of the paper is separated into the following sections after the introduction in section I: An overview of the many types of RFID principles is provided in Section II. The working of RFID is described in section III. Section IV provides a description of the proposed model. Finally, the final section presents the conclusions.

II. RFID Principles

Although there are varied types of RFID, we may divide them into two groups at the most fundamental level: active and passive. The fundamental RFID design methodologies for delivering energy from source (the reader) to the receiver (the tag) are: electromagnetic (EM) wave capture and magnetic induction. These two systems make use of an RF antenna's near-field and far-field electromagnetic properties [12].

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A. *Near Field RFID*

Near-field technology is derived from the concept of Faraday about magnetic induction. If a reading coil gets a large alternating current, an alternating magnetic field is created around that coil. A tag with a smaller coil (Fig. 1) may be used in this field and will function [16], producing an alternating voltage across it. A charge reservoir formed when this voltage is rectified and connected to a capacitor and may then be utilized to supply the tag chip. Load modulation is used by near-field coupling tags to return data to the reader. A little magnetic field is created around any extracted current, which opposes the magnetic field of the reader, thus the reader coil may detect this as a modest rise in current passing through it. The load on the tag's coil affects the current in direct proportion (hence load modulation). Although the main and secondary coils of a transformer are normally twisted together to promote effective power transfer, power transformers used in most houses today also utilize this principle. However; although a secondary coil is on a distance, it could pick up some of the energy because of the extended magnetic field beyond the primary coil. This signal could be recovered if the changes in the current flowing through the reader coil are monitored. There are many available modulation encodings, based on the needed ID bits number, rate of data transference, and further added bits to the code to eliminate mistakes brought on by the transmission channel noise. A passive RFID system could be easily built using Near-field coupling [12]

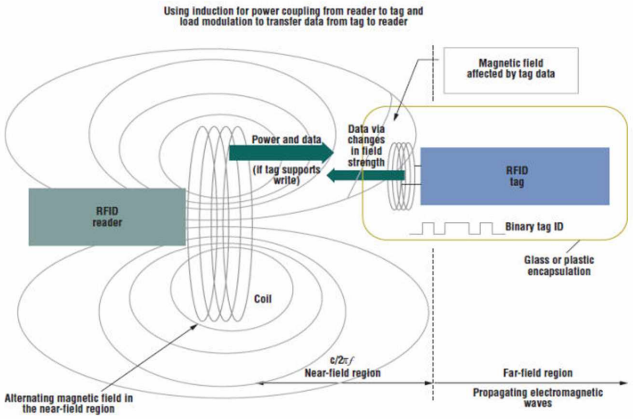


Fig. 1. power and communication system of RFID tags operating in near-field region with less than 100 MHz [12]

B. *Far Field RFID*

EM waves that travel from a reader's dipole antenna are cached by RFID tags that utilize far-field emissions. The energy is picked up by a tiny dipole antenna in the tag then is spread across the dipole arms [17]. By correcting this energy and connecting it to a capacitor, the diode may accumulate enough energy to power its circuitry. Commercial far-field RFID tag designers employ back scattering technology as shown in Fig. 2 [16]. An antenna could be able in absorbing the energy bulk at a specific frequency, which is determined in the design stage, if it is designed with neat dimensions. Also, part of tiny waves could be backed to the reader by reflection, if the is a resistance at this frequency. The reader could use a sensitive radio receiver for energy detection. Gradual adjustment of the antenna's resistance, the incoming signal may be reflected back by the tag in a form of encoding the tag ID. In reality, a transistor could be putted across dipole to detune tag of antenna and using it for this purpose to alternately switch it on and off

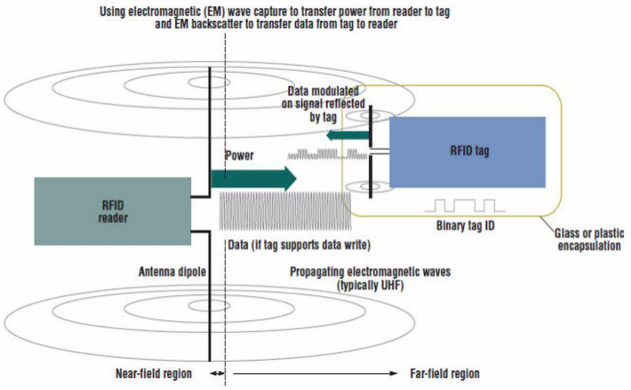


Fig. 2. power and communication system of RFID tags operating in far-field region with less than 100 MHz [12]

III. HOW RFID WORKS

An antenna and a transceiver (transmitter/receiver), which are commonly merged, make up an RFID reader. An antenna and a transponder (transmitter/responder) make up an RFID tag [17]. After delivering a radio signal by the reader to the transponder, which activates and returns data to the transceiver, the RFID tag is scanned. The main three parts that make RFID are: a coil (or antenna), a transceiver that includes decoder and an electronic programmed transponder (RF tag) with particular data.

In RFID system, data is transferred through a tag, a mobile apparatus that can achieve the requirements of a specific application by scanning and processing data. The information provided by the tag may contain identification or geographic data as well as facts about the tagged item, such as its cost, color, date of purchase, and so on. For at least ten years, thousands of companies have been utilizing RFID technology. Because of its capacity to track moving objects, RFID soon gained popularity. RFID tags will be used in increasingly pervasive and intrusive ways as the technology improves [12].

A microchip is coupled to a radio antenna set on a substrate in a standard RFID tag. The chip has a storage capacity of up to 2 kilobytes.

The information stored on an RFID tag must be retrieved via a reader. One or more antennas are used to make an ideal reader for sending radio waves and receiving the tag signals. Then, a computer system received the information that is sent by the reader in a digital form. A variety of procedures is automated when an object is linked with a particular ID [12] as depicted in Fig.3.

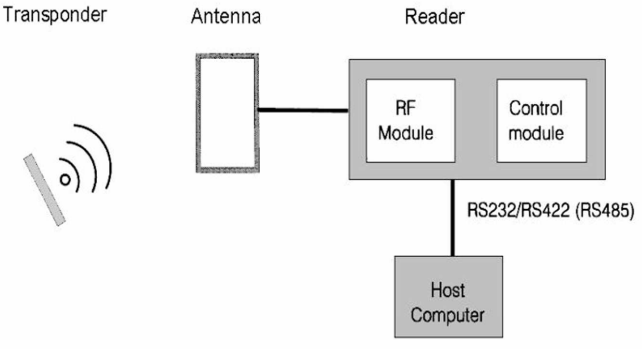


Fig. 3. Working of RFID [12]

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IV. CLOUD COMPUTING

Cloud computing is a method of computation that makes use of the Internet and virtualized shared resources. It has dynamically deployed, optimized, and recovered on-demand storage and processing resources. The main goal of cloud computing is to simplify the client's job by limiting the client's responsibilities to data input and reading while delegating more complicated tasks to "clouds." The networking computer group and data center are referred to as "clouds" [1]. By mixing a range of innovative Internet-based technologies, including as storage, computation, software, management, network, information, and other resources, cloud computing enables virtualization, on-demand customization, field deployment, and flexible mutual combination. It satisfies the user's desire for truth and provides IT services that are entirely transparent [1].

V. PROPOSED SYSTEM

We are attempting to create a smart ambulance traffic control system in our suggested system. The used hardware unites are:

- ✓ Micro-Controller (UNO)
- ✓ LED Lights (RED, YELLOW & GREEN)
- ✓ RFID Tag
- ✓ RFID Reader
- ✓ Wires

The ambulance has an RFID tag attached for location tracking, and an RFID reader is located at the signal post. RFID data is stored in the cloud system, which is accessible anytime required. The centralized system for traffic control is a microcontroller. The gathered information is processed by the microcontroller, then it changes the signal based on the result. In non-emergency cases; there is non-alert in the signal of traffic, while it changes in emergency cases after sending data to the nearest traffic post. The ambulance is registered to the cloud. The block diagram and flow chart of the proposed work is shown in Fig. 4 and Fig. 5.

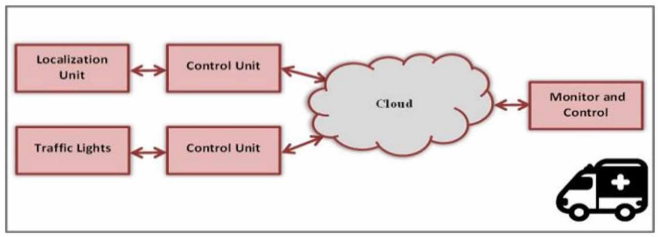


Fig. 4. Block diagram of the proposed work

A. RFID Technology

Radio Frequency Identification (RFID) is a small device that reads and records information about the tracked object using radio waves. It consists of three major components: tag and reader. In this suggested system; an RFID tag that contains three fields for data storage is installed in each ambulance. There is a distance between the reader, which may be positioned on the road or on a platform, and the traffic signal. The propagation of RF signals from and to the tagged object allows reader to read the tag. The reader delivers the data from the tag to the traffic light system after reading it

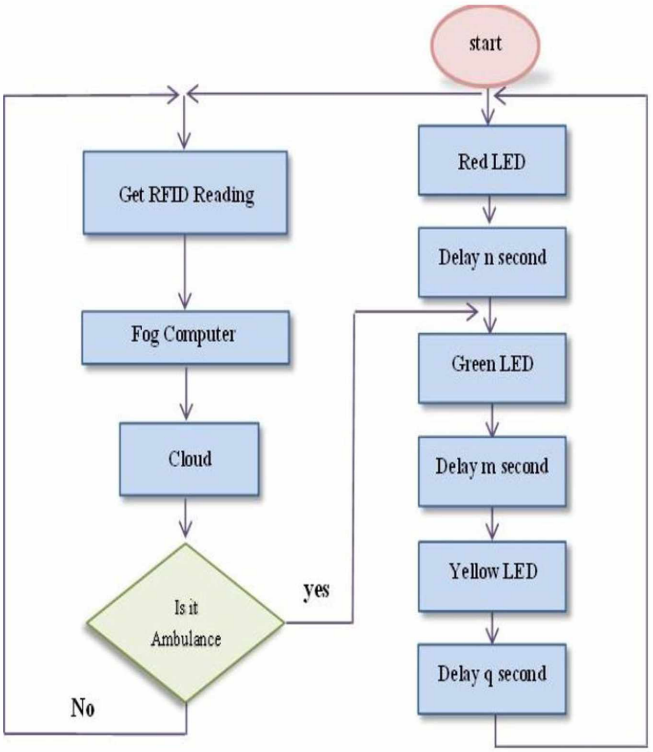


Fig. 5. Flow chart of the proposed work

B. Microcontroller

All the information of ambulance is gathered in the RFID reader, so it is necessary to be connected to the microcontroller. Here Arduino UNO microcontroller is used. The connection is depicted in Fig. 6.

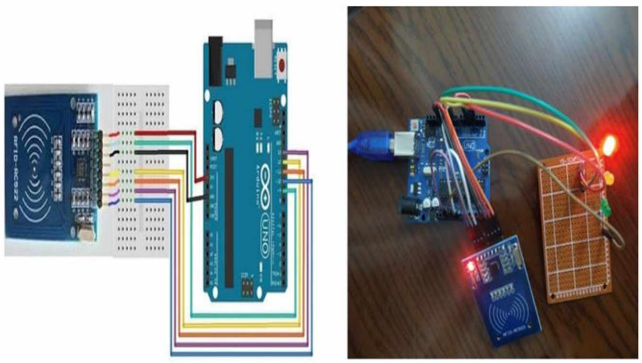


Fig. 6. Connection between RFID reader and Microcontroller

When an ambulance with RFID tag stopped by traffic, the reader delivers the data from the tag to the traffic light system after reading it. The microcontroller receives all the collected data from both the RFID and app. Then, the results are processed and the signal is modified. The microcontroller sends these data also to the Fog computer. The language that used to program the Arduino uno was #C language.

C. Fog Computer

A cable is utilized to connect fog computer with the microcontroller. As clarified in Fig.7 the numbers of RFID tag of each ambulance were displayed in computer. The fog computer sends theses data to cloud to check the ID existence for authorization. Fig. 8 depicts how the data of the cards appeared in the cloud.

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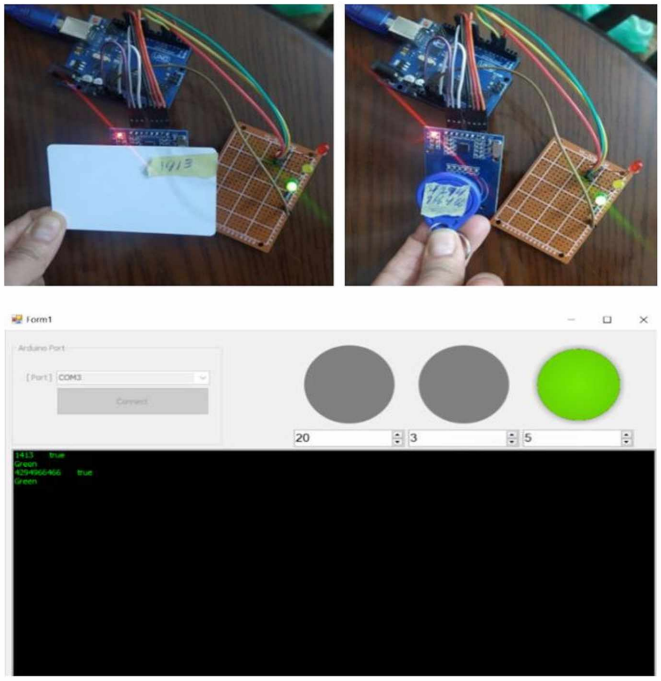


Fig. 7. The data in fog computer, RfID card numbers

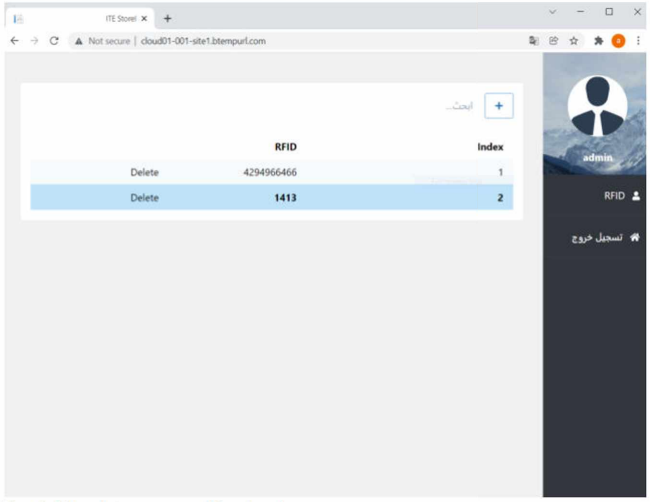


Fig. 8. The data appeared in cloud

VI. CONCLUSION

Human life is valuable, and all elements of it, including ambulance services, must adhere to strict safety standards. We may be able to provide uninterrupted traffic control system servicing by deploying intelligent ambulance systems by adding various signal change approaches to enable for flow control. Because the RFID's accuracy is higher than that of cameras, our proposed work improves traffic signal infraction detecting systems' performance. This technology is cost-effective, multi-functional, and cloud-based, making it more efficient. Because other vehicles must assemble to allow ambulance services to pass, regular accidents occurrence at traffic light passageway will be reduced if this technology is applied. This life-saving initiative should be implemented in traffic forums to benefit the general population.

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